accounts for the conservation of the angular momentum. The electrons emitted by the boron nucleus display a continuous series of energy values; this fact leads us to the assumption that a neutrino is emitted simultaneously in this case, too. The facts that the boron nucleus is really unstable and that it eventually changes into a carbon nucleus through the emission of an electron and a neutrino, point to the conclusion that the neutrino also has an angular momentum or spin value of $\hbar/2$, opposite in direction, or mathematical sign, to that of the electron, so that it compensates for the electron.

We have now gained a general view of the conclusions which can be drawn from the three conservation laws with respect to the stability of the nucleus. Our findings may be summed up briefly as follows: The nucleus of an atom will always change spontaneously into another nucleus if this process, firstly liberates energy, and secondly, is compatible with the laws of conservation of charge and angular momentum. It is true, of course, that this spontaneous transformation may occur after the lapse of a considerable period of time only, in other words, that its probability may be very small. But if either one of the conditions just mentioned is not given, the nucleus in question is a stable one.

II. NUCLEAR STRUCTURE

The conservation laws have enabled us to reach far-reaching conclusions concerning the stability of atoms, without resorting to any hypothesis about the conditions within the nucleus or the forces operative in it. Now let us attempt to formulate conclusions concerning the internal structure of the nucleus, on the ground of experiments, independently of specific assumptions about these internal forces. How are the protons and neutrons distributed within the nucleus? Can we compare a nucleus with, say, a drop of liquid, in which the molecules are packed with uniform density? Or should the nucleus be compared rather to a globular star cluster, in the centre of which the stars are extremely close to one another, but more widely separated with greater distances from the centre?

In this connection, the mass defects render a very important service. From them we can calculate the binding energy, and we find that when reckoned for individual particles within the atom, the binding energy is approximately the same in all atoms. The lighter atoms—up to about aluminium—where the absolute magnitude is smaller, are an exception to this rule. In the other atoms, the binding energy of each nuclear particle is always between 6 and 9 Mev. It seems, therefore, that all nuclear particles are bound more or less equally firmly.

We may draw a further conclusion from the size of the nucleus. We can